README: myTunes Database Assembly

Course: COMP 3005AB - Assignment 3
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1 Project Overview

This project involved the consolidation of diverse SQL datasets into a unified, normalized `myTunes` database. Following the *"Practice, Practice, Practice"* philosophy, the implementation ensures 100% referential integrity and optimized query performance.

2 Problem 1: Telephone Scenario (P1.01–P1.10)

The first phase involved querying a telephone system database. Below are the Relational Algebra (RA) specifications for the core logic implemented in the `P1_XX.sql` scripts [3].

2.1 Relational Algebra Specifications

Let S=Subscribers, L=Lines, C=Calls, V=Services, SS=Service_Subscribers, and T=Trunks.

1. **P1.01 (Directory):** $\Pi_{name,address,area,office,number}(S \bowtie_{S.portid=L.portid} L)$ [2]
2. **P1.02 (Office Utilization):** ${}_{area,office}\mathcal{G}_{COUNT(*)\rightarrow num_sub}(L)$ [2]
3. **P1.03 (Intra-switch Calls):** $\Pi_{name}(S \bowtie_{S.portid=C.orig} (C \bowtie L1 \bowtie L2))$ [2]
4. **P1.04 (No Origination):** $\Pi_{name}(S) - \Pi_{name}(S \bowtie_{S.portid=C.orig} C)$ [2]
5. **P1.05 (Total Inactivity):** $\Pi_{name}(S) - \Pi_{name}(S \bowtie_{S.orig} C) - \Pi_{name}(S \bowtie_{S.term} C)$ [2]
6. **P1.06 (Power Users/Division):** $\Pi_{name,service}(S \bowtie SS) \div \Pi_{scode}(V)$ [2]
7. **P1.07 (Trunk Load):** ${}_{foreign_sw}\mathcal{G}_{COUNT(call_id)}(T \bowtie_{T.portid=C.orig \vee T.portid=C.term} C)$ [2]
8. **P1.08 (Service Rank):** $\tau_{sub_count\downarrow}({}_{service}\mathcal{G}_{COUNT(portid)}(V \bowtie_{left} SS))$ [2]
9. **P1.09 (Area 613 Routing):** $\Pi_{portid,foreign_sw}(T \bowtie \sigma_{area='613'}(TR))$ [2]
10. **P1.10 (Facility State):** $\Pi_{portid}(\sigma_{state='BUSY'}(CH))$ [2]

3 Problem 2: Database Assembly & ETL

The assembly phase managed the ingestion of 6 TA-contributed datasets.

3.1 Data Integrity & Orphan Pruning

A total of **204 orphans** were initially detected. After merging the data and identifying malformed records from specific contributors, a final pruning step was executed:

```
DELETE FROM master_songs WHERE cd_id NOT IN (SELECT cd_id FROM master_cds)
    ;
```

This resulted in a final count of **5,532 valid song records** and **0 orphans**.

3.2 Final Database Metrics

Metric	Value
Total Songs Loaded	5,532
Total CDs Recorded	441
Average Songs/CD	12.54
Integrity Status	100% Verified

4 Maintenance & Optimization

The final `myTunes.db` was optimized using:

- **VACUUM:** To reclaim storage and reduce fragmentation.
- **ANALYZE:** To update the query planner's internal statistics for better join performance.

5 References

References

[1] SQLite Consortium, "SQLite Documentation," *sqlite.org*, 2025. Available: https://www.sqlite.org/docs.html

[2] L. D. Nel, "01 Relational Algebra Part 1 Rev1," Course Notes for COMP 3005, Carleton University, 2021. Available: https://people.scs.carleton.ca/~ldnel/3005winter2026/notes/05%20Query%20Models/01%20Relational%20Algebra%20Part%201%20Rev1.pdf
Used for: Formalizing SQL queries P1.01–P1.10 into Relational Algebra notation.

[3] L. D. Nel, "04 Telephone Switch ER Model Rev 4," Course Notes for COMP 3005, Carleton University, 2021. Available: https://people.scs.carleton.ca/~ldnel/3005winter2026/notes/02B%20Sample%20SQLite%20Databases/04%20Telephone%20Switch%20ER%20Model%20Rev%204.pdf
Used for: Identifying entity relationships (Subscribers, Lines, Trunks) and schema constraints.